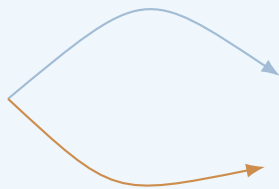


Physics Lectures

Classical Field Theory

Lecture Notes

Fields, variational principles, symmetries, and conservation laws.



June 9, 2026

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Preface

Purpose and Scope

These notes are a working book on classical field theory. The central idea is to treat fields as physical degrees of freedom distributed over space and time, and to derive their equations from variational principles.

The intended path is gradual. We begin with the meaning of a field, then move to Lagrangian field theory, symmetries, conservation laws, and standard examples such as scalar fields, electromagnetism, and continuum models. Later chapters can connect the same language to relativistic notation and gauge theory.

Remark

This book should be developed chapter by chapter. Each new topic should be added as a small source file, compiled, reviewed in the PDF, and then improved in the next step.

Chapter 1

Fields as Physical Objects

1.1 What Is a Field?

A field is an assignment of physical data to each point of a region of space, spacetime, or another underlying domain. Instead of describing a system by finitely many coordinates $q^1(t), \dots, q^n(t)$, field theory describes a system by functions such as

$$\phi(t, \mathbf{x}), \quad \mathbf{E}(t, \mathbf{x}), \quad g_{\mu\nu}(x).$$

The value of the field may be a number, a vector, a tensor, or a more elaborate geometric object.

The important conceptual change is that the degrees of freedom are distributed. At every point $\mathbf{x}$, a scalar field $\phi(t, \mathbf{x})$ has a value. The collection of all these values forms the state of the field at time t .

Definition

A *classical field* is a function or geometric assignment whose value is specified at each point of a continuous domain. In a nonrelativistic setting the domain is often space and time separately. In a relativistic setting the domain is spacetime.

Classical field theory studies the equations that determine how these assignments vary and interact.

1.2 Examples of Fields

The simplest example is a scalar field

$$\phi : \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R},$$

where $\phi(t, \mathbf{x})$ assigns one real number to each point of space at each time. Temperature in a material body is a familiar nonrelativistic scalar field.

A vector field assigns a vector to each point. The velocity field of a fluid is written

$$\mathbf{v}(t, \mathbf{x}) = (v^1(t, \mathbf{x}), v^2(t, \mathbf{x}), v^3(t, \mathbf{x})).$$

It describes the local velocity of the material element passing through $\mathbf{x}$ at time t .

Electromagnetism is naturally expressed in terms of fields. In elementary vector notation one uses the electric and magnetic fields

$$\mathbf{E}(t, \mathbf{x}), \quad \mathbf{B}(t, \mathbf{x}).$$

In relativistic notation these are combined into the electromagnetic field strength tensor $F_{\mu\nu}$.

Example

At a fixed time, the temperature inside a room is not one number. It is a function $T(\mathbf{x})$. Heating, cooling, and diffusion are statements about how this function changes in time and space.

1.3 Locality and Degrees of Freedom

Most classical field theories are local. This means that the equation at a point depends on the value of the field and finitely many derivatives of the field at that point. For a scalar field, a typical local equation may involve

$$\phi, \quad \frac{\partial \phi}{\partial t}, \quad \nabla \phi, \quad \frac{\partial^2 \phi}{\partial t^2}, \quad \Delta \phi.$$

It does not directly require the value of ϕ at every distant point.

This locality principle is one reason differential equations are central in field theory. Partial derivatives compare nearby field values, so local physical laws become partial differential equations.

Summary

A finite-dimensional mechanical system has coordinates $q^i(t)$. A field has degrees of freedom labelled by position as well as by time, such as $\phi(t, \mathbf{x})$. Classical field theory is therefore the natural language for waves, fluids, electromagnetic fields, and relativistic continuous systems.

Chapter 2

Lagrangian Field Theory

2.1 From Mechanics to Fields

In analytical mechanics, the motion of a system is described by generalized coordinates $q^i(t)$. A Lagrangian $L(q, \dot{q}, t)$ defines the action

$$S[q] = \int_{t_0}^{t_1} L(q, \dot{q}, t) dt.$$

The physical path is obtained by requiring the first variation of the action to vanish.

Field theory uses the same principle, but the unknown is now a field $\phi(t, \mathbf{x})$. The role of the Lagrangian is played by a Lagrangian density $\mathcal{L}$, because the action must be integrated over space and time.

Remark

The word density matters. In mechanics the Lagrangian is integrated over time. In field theory the Lagrangian density is integrated over spacetime volume.

2.2 The Action Functional

For one real scalar field in flat spacetime, a common form of the action is

$$S[\phi] = \int \mathcal{L}(\phi, \partial_\mu \phi, x) d^4x.$$

Here $\partial_\mu \phi$ denotes the first partial derivatives of the field, and d^4x denotes the spacetime volume element in the chosen coordinates.

The action is a functional: it assigns a number to an entire field configuration. The variational principle states that the physical field configuration makes the action stationary under small variations that vanish on the boundary.

Definition

An *action functional* in classical field theory is a map

$$\phi \mapsto S[\phi]$$

from field configurations to real numbers, usually written as an integral of a local Lagrangian density.

2.3 Euler–Lagrange Equations for a Scalar Field

Suppose the Lagrangian density depends on a scalar field and its first derivatives:

$$\mathcal{L} = \mathcal{L}(\phi, \partial_\mu \phi, x).$$

The stationary action condition gives the field equation

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) = 0.}$$

This is the field-theoretic Euler–Lagrange equation.

For several fields ϕ^a , there is one equation for each field component:

$$\frac{\partial \mathcal{L}}{\partial \phi^a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^a)} \right) = 0.$$

The result is usually a partial differential equation rather than an ordinary differential equation.

Chapter 3

Symmetries and Conservation Laws

3.1 Why Symmetries Matter

A symmetry is a transformation that changes the description of a system without changing the physical content of the action. In field theory, symmetries are not only aesthetic. They determine conserved quantities and strongly restrict the form of possible field equations.

Spatial translation symmetry is related to momentum conservation. Time translation symmetry is related to energy conservation. Rotational symmetry is related to angular momentum conservation. Internal symmetries lead to conserved charges.

The power of the Lagrangian formulation is that these connections can be stated systematically.

3.2 Preview of Noether's Theorem

Noether's theorem says that continuous symmetries of the action produce conserved currents. In field theory this means that a symmetry leads to a current j^μ satisfying a local conservation law

$$\partial_\mu j^\mu = 0.$$

This equation expresses conservation locally. For example, it can imply that a total charge

$$Q(t) = \int_{\mathbb{R}^3} j^0(t, \mathbf{x}) \, d^3x$$

is independent of time, provided suitable boundary conditions hold.

Summary

Classical field theory is built from fields, actions, variational principles, and symmetries. Noether's theorem is the bridge between symmetry and conservation.

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